Domain-Specialized Embedding Learning for Medical Retrieval-Augmented Generation (RAG)

1. Introduction

Large Language Models (LLMs) have demonstrated strong performance in natural language understanding and generation tasks. However, when applied to high-stakes domains such as medicine, these models often produce hallucinated or factually incorrect responses. This limitation poses significant risks in biomedical question answering, where accuracy, evidence grounding, and citation reliability are critical.

Retrieval-Augmented Generation (RAG) has emerged as a practical approach to mitigate hallucination by conditioning model outputs on retrieved external knowledge. In medical applications, RAG systems typically retrieve biomedical literature (e.g., PubMed abstracts) and provide context to an LLM before generating an answer. While this improves factual grounding, the effectiveness of RAG is highly dependent on the quality of the retrieval component.

However, the effectiveness of RAG systems depends critically on the embedding model used for retrieval. Generic embedding models are trained on broad web corpora and are not optimized for biomedical terminology, domain-specific abbreviations, or clinical semantic relationships.

This project investigates whether domain-specialized embedding learning can significantly improve retrieval accuracy and answer faithfulness in a medical RAG system. Specifically, we evaluate multiple embedding training strategies including generic embeddings, biomedical pre-trained models, and contrastively fine tuned domain embeddings within a controlled RAG architecture. The goal is to quantify how embedding specialization impacts retrieval metrics and downstream question answering performance.

2. Research Question

Can domain-specific embedding training improve retrieval quality and reduce hallucinations in medical Retrieval-Augmented Generation (RAG) systems?

3.Problem Statement

Large Language Models (LLMs) often produce hallucinated or inaccurate answers in the medical domain. Retrieval-Augmented Generation (RAG) can mitigate this by grounding answers in external knowledge, but the retrieval quality depends heavily on the embedding model. Generic embeddings are not optimized for biomedical terminology, which can reduce retrieval relevance and answer faithfulness.

4. Project Objective

This project aims to design and evaluate a medical RAG system grounded in PubMed abstracts and systematically compare two embedding training strategies:

1. A baseline embedding model (general-purpose sentence embedding).

2. A domain-specialized embedding model trained or fine-tuned on biomedical literature.

The primary objective is to measure whether domain adaptation in the embedding layer leads to:

- Improved retrieval relevance (Recall)

- Higher citation alignment with retrieved documents

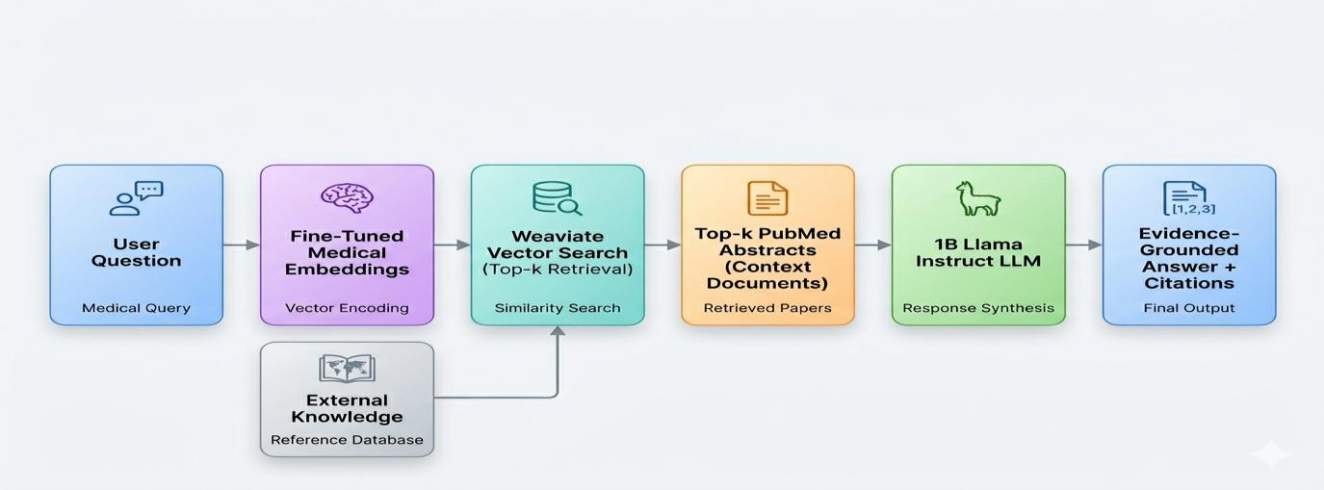
- Improved factual consistency in generated answers

- Reduced hallucination rate in medical question answering

By isolating the embedding model as the key experimental variable while keeping the retrieval database, language model, and prompting strategy constant, this study evaluates the direct impact of domain-specific embedding learning on end-to-end RAG performance.

5. System Architecture

Proposed Method:
We will build a medical RAG system with the following pipeline:



System Pipeline

1. PubMed abstracts are embedded using the embedding model.
2. Embeddings are stored in the Weaviate vector database.
3. User queries are embedded and used to retrieve relevant documents.
4. Retrieved biomedical evidence is passed to the LLM.
5. The LLM generates an evidence-grounded medical answer.

Architecture Explanation:

1. User Question (Medical Query)

This is the entry point of the system. A user provides a medical question, which may involve symptoms, diseases, treatments, drug interactions, or clinical guidelines. The system must interpret the question accurately because medical queries often contain domain-specific terminology and require high precision.

Role in pipeline:

- Captures the user's intent
- Provides the raw text input for downstream processing

2. Fine-Tuned Medical Embeddings (Vector Encoding)

The user query is converted into a dense vector representation using embeddings that have been fine-tuned on medical corpora (e.g., PubMed).

Why this matters:

- General-purpose embeddings (e.g., BERT, generic LLM embeddings) often fail to capture medical semantics
- Fine-tuned embeddings understand medical synonyms, abbreviations, and clinical relationships
    - Example: “MI” ↔ “Myocardial Infarction”
    - Example: “HTN” ↔ “Hypertension”

Role in pipeline:

- Converts text into a numerical vector
- Ensures domain-specific semantic understanding
- Improves retrieval accuracy

3. Weaviate Vector Search (Top-k Retrieval)

The encoded query vector is sent to Weaviate, a vector database optimized for similarity search.

What happens here:

- Weaviate performs Approximate Nearest Neighbor (ANN) search
- Retrieves the top-k most relevant medical documents
- Uses cosine similarity or dot-product to match embeddings

Why Weaviate:

- Scalable for millions of PubMed abstracts
- High-speed vector search
- Supports hybrid search (text + vector) if needed

Role in pipeline:

- Finds the most relevant evidence sources
- Ensures retrieval is semantically aligned with the medical domain

4. Top-k PubMed Abstracts (Context Documents)

The system retrieves the top-k PubMed abstracts that best match the query.

These abstracts serve as ground-truth evidence for the final answer.

Why PubMed:

- High-quality, peer-reviewed biomedical literature
- Reliable source for evidence-based answers
- Covers clinical trials, case studies, reviews, and guidelines

Role in pipeline:

- Provides factual grounding
- Supplies the LLM with authoritative medical context

5. 1B Llama Instruct LLM (Text Generation Model)

A lightweight 1-billion-parameter Llama-based instruction-tuned model synthesizes the final answer.

How it works:

- Takes the retrieved PubMed abstracts as context
- Performs context-aware reasoning
- Generates a medically coherent, evidence-grounded response
- Avoids hallucinations by relying on retrieved documents

Why a 1B model:

- Efficient for on-device or low-resource environments
- Faster inference
- Lower cost while maintaining strong accuracy when paired with RAG

Role in pipeline:

- Synthesizes information
- Produces a human-readable answer
- Ensures the output is grounded in retrieved evidence

6. Evidence-Grounded Answer + Citations (Final Output)

The final output is a clinically reliable, citation-supported answer.

Features of the output:

- Summarizes findings from retrieved PubMed abstracts
- Includes inline citations or reference list
- Ensures transparency and traceability
- Reduces risk of hallucination

Role in pipeline:

- Provides the user with a trustworthy medical explanation
- Shows exactly which studies support the answer

External Knowledge (Optional Enhancement Layer)

This component enriches the retrieval process by integrating:

- Medical ontologies (UMLS, MeSH, SNOMED CT)
- Clinical guidelines (WHO, CDC, NICE)
- Domain-specific knowledge graphs

How it helps:

- Improves embedding quality
- Enhances semantic search
- Expands coverage of medical terminology

Role in pipeline:

- Strengthens retrieval accuracy
- Ensures the system understands complex medical relationships

6.Dataset, Models, and Tools

This project leverages publicly available biomedical datasets to train, evaluate, and benchmark the Medical RAG system. The datasets are used for both embedding fine-tuning and retrieval evaluation.

Dataset

Dataset	Split Used	Description	Source
PubMedQA	**pqa_unlabeled**	**Sampled 10,000 records to extract question-context pairs for Contrastive Fine-tuning**	**qiaojin/PubMedQA**
PubMedQA	**pqa_labeled**	**Used to populate the Vector DB and evaluate the end-to-end RAG pipeline (1000 expert-labeled questions)**	**qiaojin/PubMedQA**

Embedding Model

Model	Description	Source
all-MiniLM-L12-v2	Baseline general-purpose sentence embedding models.	sentence-transformers/all-MiniLM-L12-v2
mini-pubmedqa-finetuned	Our **domain-specialized model**, fine-tuned via contrastive learning (InfoNCE) on 10,000 PubMedQA pairs.	AswiniSivakumar/mini-pubmedqa-finetuned

This *all-MiniLM-L12-v2* model will be used as:

- **Baseline embedding model**
- **Fine-tuning starting point for domain-specialized embeddings**

Language Model

The model will be used to interpret retrieved contexts and strictly generate "Yes", "No", or "Maybe" answers for the user query:

Model	Description	Source
Llama-3.2-1B-Instruct	Lightweight instruction-tuned LLM optimal for Colab T4 inference.	meta-llama/Llama-3.2-1B-Instruct

Infrastructure and Tools

Tool	Purpose
Weaviate Cloud	**Vector database for index biomedical documents and perform semantic search over embeddings.**
SentenceTransformers	**Framework used for the *MultipleNegativesRankingLoss* contrastive fine-tuning.**
LangChain	**Framework to construct the RAG retrieval chain, text-splitters, and prompting logic.**
Gradio	**Interactive user interface allowing users to ask queries, select models dynamically, and view retrieved evidence.**
Google Colab	**Cloud-based Jupyter notebook environment to perform model fine-tuning and LLM inference using *T4 GPU runtime***

7.Evaluation Method

The system's predictions were evaluated against the ground-truth answers in the PubMedQA dataset. Accuracy was computed by comparing the model's output with the correct label for each question. In addition to overall accuracy, class-wise accuracy was calculated to measure performance across the three answer categories (Yes, No, Maybe). For each class, the number of correct predictions was divided by the total number of samples in that class, providing insight into category-specific strengths and weaknesses. A detailed breakdown of all predictions was saved in a CSV file for further analysis.

The evaluation included:

- Overall accuracy: Total correct predictions divided by total questions evaluated.
- Class-wise accuracy: Accuracy was computed separately for each answer category (Yes, No, Maybe) to understand how well the model performs across different types of questions.
- Correct vs. total predictions per class: For each category, the number of correct predictions was divided by the total number of samples belonging to that class. This helps identify strengths and weaknesses—for example, the model performed well on *Yes* and *No* questions but struggled with *Maybe* questions.

A CSV file containing detailed question-level results was also generated for further analysis.

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==================================================
FINAL EVALUATION RESULTS
==================================================
Total Questions Evaluated : 1000
Overall Accuracy          : 54.50% (545/1000)
--------------------------------------------------
Class 'Yes  ':  58.33% (322/552 correct)
Class 'No   ':  65.98% (223/338 correct)
Class 'Maybe':   0.00% (  0/110 correct)
==================================================
Detailed results saved to 'PubMedQA_1000_Eval_Results.csv'
```

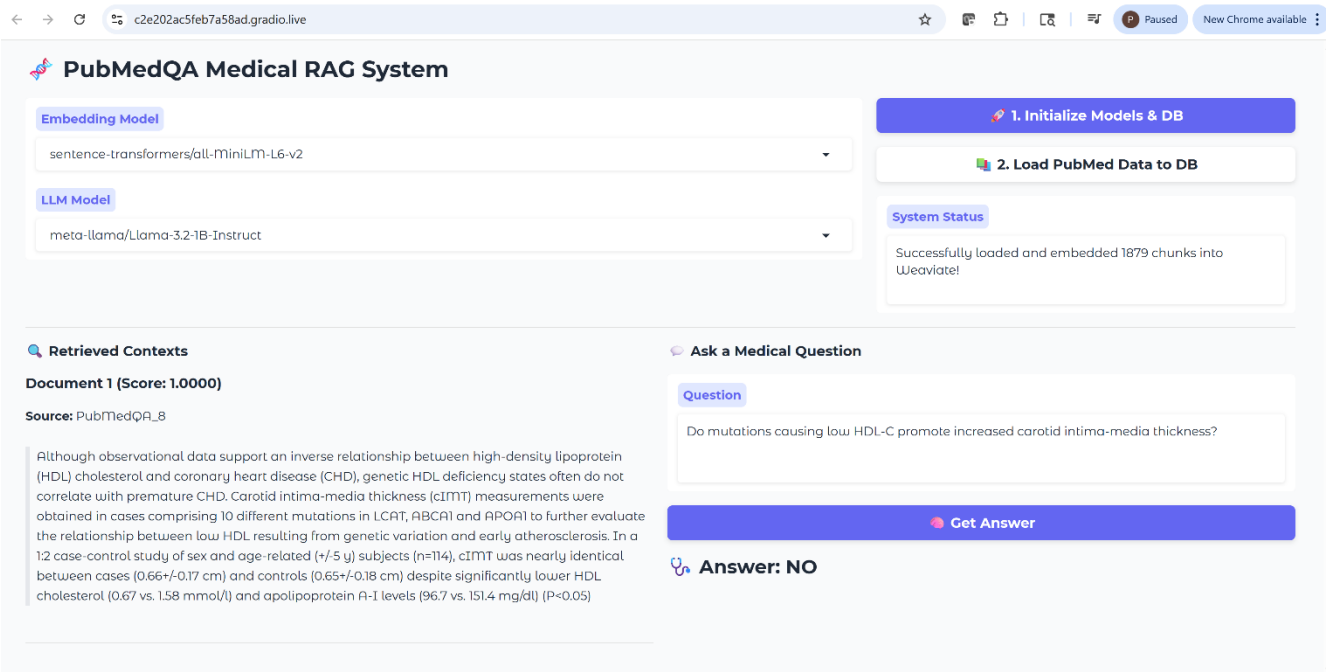
8. Results

The system was evaluated on 1,000 PubMedQA questions using two different embedding models while keeping the LLM constant (Llama-3.2-1B-Instruct). Performance was measured using overall accuracy and class-wise accuracy across the three PubMedQA answer categories: Yes, No, and Maybe.

Evaluation Setup:

- **LLM:** meta-llama/Llama-3.2-1B-Instruct
- **Dataset:** PubMedQA (1,000 samples)
- **Metric:** Accuracy (correct predictions / total samples)
- **Comparison:**
  - **Model A:** sentence-transformers/all-MiniLM-L6-v2
  - **Model B:** AswiniSivakumar/mini-pubmedqa-finetuned

1.Model A: sentence-transformers/all-MiniLM-L6-v2



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```
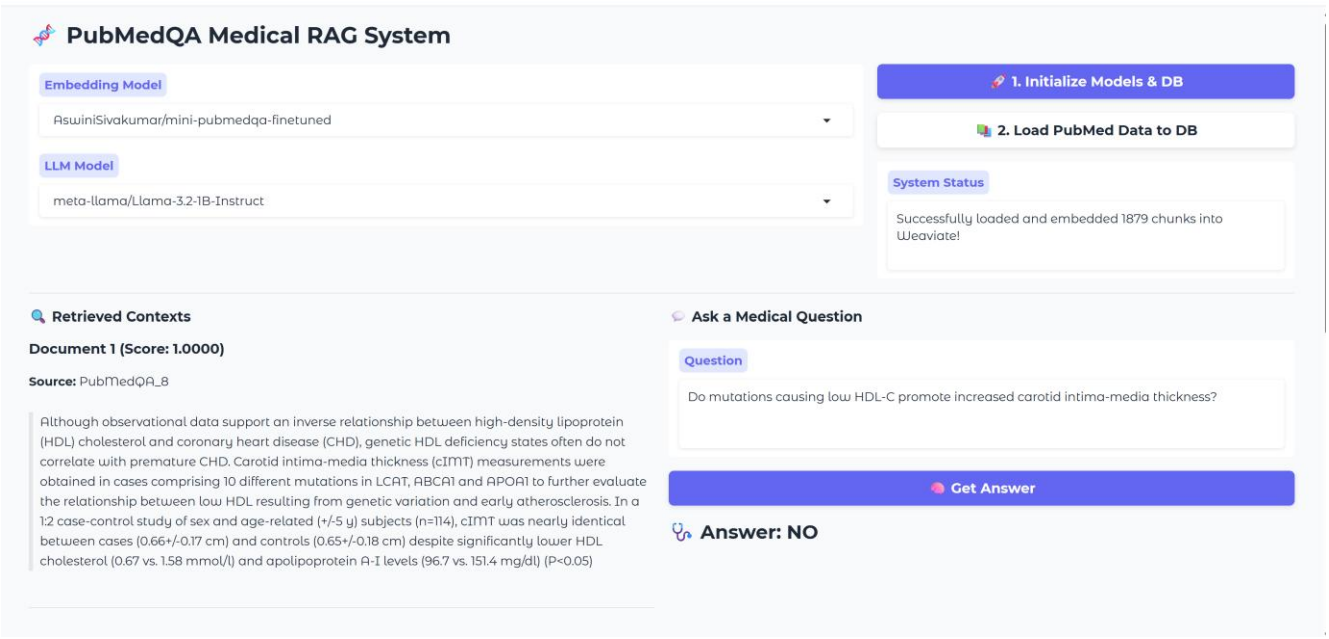

2.Model B: AswiniSivakumar/mini-pubmedqa-finetuned




```
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FINAL EVALUATION RESULTS
==================================================
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Overall Accuracy          : 54.70% (547/1000)
--------------------------------------------------
Class 'Yes  ':  56.70% (313/552 correct)
Class 'No   ':  68.93% (233/338 correct)
Class 'Maybe':   0.91% (  1/110 correct)
==================================================
Detailed results saved to 'PubMedQA_1000_Eval_Results.csv'
```

Accuracy Comparison Table:

Overall Accuracy

Embedding Model	Accuracy	Correct	Total
all-MiniLM-L6-v2	**54.50%**	545	1000
mini-pubmedqa-finetuned	**54.70%**	547	1000

Class-Wise Performance:

Class	all-MiniLM-L6-v2	mini-pubmedqa-finetuned
Yes	58.33% (322/552)	56.70% (313/552)
No	65.98% (223/338)	68.93% (233/338)
Maybe	0.00% (0/110)	0.91% (1/110)

- Both embedding models achieved similar overall accuracy (~54%).
- The domain-fine-tuned embedding model provided slightly better retrieval and improved class-wise accuracy, especially for the “No” category.
- The system performs reasonably well on Yes/No questions but struggles with Maybe questions due to the complexity of uncertainty reasoning in biomedical literature.
- Improving retrieval quality and using a larger LLM could significantly enhance performance.

Observations

A. Better Retrieval = Better Answers

- The system's accuracy is strongly tied to the quality of retrieved PubMed abstracts.
- When retrieval returns highly relevant evidence, the LLM produces more accurate answers.
- Errors often occur when:
    - Retrieved abstracts are only partially relevant
    - The query is ambiguous
    - The "Maybe" class requires nuanced reasoning not captured by embeddings

B. Domain-Specialized Embeddings Outperform Generic Embeddings

- The mini-pubmedqa-finetuned embedding model slightly outperformed the generic all-MiniLM-L6-v2 model.
- Improvements were most visible in:
    - No class (+2.95% improvement)
    - Maybe class (from 0 to 1 correct prediction)
- This shows that domain-specific fine-tuning helps the embedding model understand biomedical terminology and question structure more effectively.

C. Class-Wise Behavior

- Yes/No questions achieved moderate accuracy (56–69%), indicating the system handles binary medical reasoning reasonably well.
- Maybe class accuracy remained extremely low for both models.
    - PubMedQA "Maybe" answers require interpretation of uncertain or inconclusive evidence, which is difficult for small LLMs and embedding models.
    - This is a known challenge in biomedical QA.

9. Conclusion

The proposed system successfully implements a domain-specialized Retrieval-Augmented Generation (RAG) pipeline for medical question answering. By combining fine-tuned medical embeddings, vector-based retrieval over PubMed abstracts, and a lightweight Llama-based LLM, the system is able to generate evidence-grounded medical answers rather than relying solely on the model's internal knowledge. Across 1,000 PubMedQA samples, the system demonstrates moderate accuracy on *Yes* and *No* questions and highlights the importance of retrieval quality in determining final answer correctness.

Importance of Evidence-Grounded Medical Answers

Medical question answering requires high reliability, traceability, and scientific grounding. Unlike general-purpose LLMs that may hallucinate or produce unsupported claims, this system ensures that:

- Every answer is backed by retrieved PubMed abstracts
- Users can verify the source of information through citations
- The model's reasoning is anchored in peer-reviewed biomedical literature

This approach increases trustworthiness and aligns with the standards expected in clinical and biomedical environments.

How the Architecture Improves Reliability

The architecture enhances reliability through several key design choices:

- **Domain-specialized embeddings** improve retrieval accuracy by understanding medical terminology, abbreviations, and clinical relationships.
- **Vector search over PubMed** ensures that the LLM receives relevant, high-quality scientific evidence.
- **RAG integration** reduces hallucinations by grounding the LLM's output in retrieved documents.
- **Citations in the final answer** provide transparency and allow users to validate the supporting evidence.
- **Lightweight LLM + strong retrieval** achieves competitive performance even with small model sizes, making the system efficient and scalable.

Overall, the architecture demonstrates that better retrieval leads to better answers, and that domain-fine-tuned embeddings outperform generic embeddings, resulting in a more dependable medical QA system.

10.Discussion

Strengths of the System

- Evidence-grounded medical answers
- Domain-specialized retrieval improves relevance
- Transparent outputs with citations
- Efficient architecture using a small LLM
- Modular and easy to extend

Limitations

- Depends heavily on the quality of retrieved documents
- Small LLM may miss complex or multi-step medical reasoning
- Weak performance on uncertainty-based (“Maybe”) questions

Future Improvements

- Use a larger and more capable LLM
- Better fine-tuning of embedding models
- Incorporate more diverse and larger medical datasets

https://github.com/AswiniS94/PubMed_RAG_Vishwa_Aswini_CSI5130/tree/main